KIGALI INDEPENDENT UNIVERSITY (ULK)
DEPARTMENT OF COMPUTER SCIENCE / IT
COURSE: UML CLASS DIAGRAM MODELLING
INSTRUCTOR: ERIC MANIRAGUHA

REPORT 1: PROJECT DESCRIPTION REPORT

Project Title: Outpatient Clinic Management System (OCMS)
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Date: October 2026

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1. BUSINESS IDEA
The Outpatient Clinic Management System (OCMS) is designed for healthcare clinics and outpatient centers to automate the daily management of patient visits, doctor schedules, consultation histories, and billing operations.

2. PROBLEM STATEMENT
Many outpatient clinics still rely on manual paper files to schedule visits and track patient histories. This traditional process causes frequent doctor double-booking, lost or damaged patient records, delays in consultation, and errors during billing and invoicing.

3. PURPOSE OF THE PROPOSED SYSTEM
The purpose of the system is to provide a structured, object-oriented software model that:
- Centralizes patient registration and doctor profile management.
- Streamlines the scheduling, updating, and cancelling of appointments.
- Maintains a permanent, organized digital medical history for every patient.
- Automatically creates consultation bills upon service delivery.

4. PROJECT SCOPE

4.1 Functions Covered (In Scope):
- Managing patient demographic information and doctor specializations.
- Scheduling and tracking appointment statuses (e.g., Scheduled, Completed, Cancelled).
- Documenting doctor diagnoses, clinical observations, and prescriptions.
- Generating consultation invoices and tracking payment status.

4.2 Functions Excluded (Out of Scope):
- Inpatient ward admissions and hospital bed allocations.
- Pharmacy warehouse inventory tracking and supplier purchases.
- Laboratory diagnostic equipment and blood-bank operations.

- Complex third-party health insurance claim processing.

5. MAIN CLASSES AND RESPONSIBILITIES
- Person: Base parent class holding shared demographic attributes (name, phone, email).
- Patient: Represents the clinic client; books appointments and views medical history.
- Doctor: Represents the medical practitioner; conducts appointments and writes prescriptions.
- Appointment: Manages the scheduling time, date, and status connecting doctor and patient.
- MedicalRecord: Securely holds diagnostic findings and prescription notes for a patient.
- Bill: Calculates fees and tracks payment settlement for completed appointments.

6. SYSTEM REQUIREMENTS
- Functional Requirements:
  1. The system shall allow new patients and doctors to be registered with unique IDs.
  2. The system shall enable patients to schedule an appointment with an available doctor.
  3. The system shall allow doctors to record diagnoses and prescribe medication.
  4. The system shall generate a bill automatically upon consultation completion.
- Non-Functional Requirements:
  1. Data Integrity: A medical record must permanently remain linked to its patient.
  2. Modularity: The architecture must allow future expansion (e.g., adding lab or pharmacy modules).